

Supplementary results: A Causal Evaluation of PCA-Guided Sensor-Rate Allocation under Hard Communication Budgets

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All entries are mean $\pm$ sample standard deviation over five joint experimental seeds (42, 123, 456, 789, 1024). These seeds vary acquisition, detector randomness, and the Isolation Forest fitting subset. Full Data is a full-rate reference and is listed only once per protocol. TEP point and event metrics here use binary detection (any fault). Event boundaries never cross simulation runs. Delay is in samples and conditional on detection; missed events are reported separately. A dash denotes an undefined delay. Payload bytes assume four bytes per retained value and exclude network overhead. RMSE and MAE use train-standardized values. NRMSE divides channel RMSE by the pooled test range, using one for a constant channel; those ranges are used only for reporting. The main paper uses threshold quantile 0.99. Other predeclared quantiles are sensitivity results and do not select the primary configuration.

Table 1: SMD: method retrained. Detection metrics; delay is conditional and measured in samples.

Dataset	Method	Budget	Precision	Recall	FPR	Event recall	Missed events	Delay (samples)	FA/1000	Point F1	Specificity	Events
SMD-1-1	PCA-Triage	0.3	0.318 $\pm$ 0.009	0.526 $\pm$ 0.050	0.118 $\pm$ 0.009	0.675 $\pm$ 0.112	2.6 $\pm$ 0.9	37.114 $\pm$ 22.254	5.141 $\pm$ 0.413	0.396 $\pm$ 0.020	0.882 $\pm$ 0.009	8.0 $\pm$ 0.0
SMD-1-1	Variance	0.3	0.326 $\pm$ 0.013	0.540 $\pm$ 0.026	0.117 $\pm$ 0.012	0.700 $\pm$ 0.068	2.4 $\pm$ 0.5	21.933 $\pm$ 9.999	5.780 $\pm$ 0.876	0.406 $\pm$ 0.008	0.883 $\pm$ 0.012	8.0 $\pm$ 0.0
SMD-1-1	Threshold	0.3	0.321 $\pm$ 0.015	0.523 $\pm$ 0.023	0.116 $\pm$ 0.004	0.775 $\pm$ 0.105	1.8 $\pm$ 0.8	26.102 $\pm$ 13.551	4.797 $\pm$ 0.438	0.398 $\pm$ 0.018	0.884 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Uniform	0.3	0.319 $\pm$ 0.012	0.532 $\pm$ 0.032	0.119 $\pm$ 0.002	0.725 $\pm$ 0.163	2.2 $\pm$ 1.3	22.172 $\pm$ 15.011	4.010 $\pm$ 0.366	0.399 $\pm$ 0.018	0.881 $\pm$ 0.002	8.0 $\pm$ 0.0
SMD-1-1	PCA-Triage	0.5	0.317 $\pm$ 0.013	0.507 $\pm$ 0.023	0.114 $\pm$ 0.003	0.700 $\pm$ 0.168	2.4 $\pm$ 1.3	26.070 $\pm$ 17.873	6.159 $\pm$ 0.543	0.390 $\pm$ 0.017	0.886 $\pm$ 0.003	8.0 $\pm$ 0.0
SMD-1-1	Variance	0.5	0.329 $\pm$ 0.016	0.535 $\pm$ 0.033	0.114 $\pm$ 0.008	0.825 $\pm$ 0.143	1.4 $\pm$ 1.1	14.523 $\pm$ 15.970	6.988 $\pm$ 0.456	0.407 $\pm$ 0.019	0.886 $\pm$ 0.008	8.0 $\pm$ 0.0
SMD-1-1	Threshold	0.5	0.320 $\pm$ 0.014	0.509 $\pm$ 0.017	0.113 $\pm$ 0.007	0.725 $\pm$ 0.163	2.2 $\pm$ 1.3	28.300 $\pm$ 17.448	6.342 $\pm$ 1.036	0.393 $\pm$ 0.013	0.887 $\pm$ 0.007	8.0 $\pm$ 0.0
SMD-1-1	Uniform	0.5	0.319 $\pm$ 0.008	0.518 $\pm$ 0.009	0.116 $\pm$ 0.003	0.850 $\pm$ 0.137	1.2 $\pm$ 1.1	10.901 $\pm$ 7.975	5.337 $\pm$ 0.234	0.395 $\pm$ 0.008	0.884 $\pm$ 0.003	8.0 $\pm$ 0.0
SMD-1-1	PCA-Triage	0.7	0.318 $\pm$ 0.019	0.525 $\pm$ 0.034	0.118 $\pm$ 0.007	0.750 $\pm$ 0.153	2.0 $\pm$ 1.2	18.590 $\pm$ 12.642	7.992 $\pm$ 0.832	0.396 $\pm$ 0.023	0.882 $\pm$ 0.007	8.0 $\pm$ 0.0
SMD-1-1	Variance	0.7	0.320 $\pm$ 0.014	0.529 $\pm$ 0.027	0.118 $\pm$ 0.007	0.875 $\pm$ 0.177	1.0 $\pm$ 1.4	8.527 $\pm$ 5.246	8.680 $\pm$ 0.574	0.398 $\pm$ 0.015	0.882 $\pm$ 0.007	8.0 $\pm$ 0.0
SMD-1-1	Threshold	0.7	0.328 $\pm$ 0.010	0.544 $\pm$ 0.010	0.116 $\pm$ 0.004	0.950 $\pm$ 0.112	0.4 $\pm$ 0.9	4.358 $\pm$ 0.828	8.554 $\pm$ 0.693	0.409 $\pm$ 0.010	0.884 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Uniform	0.7	0.320 $\pm$ 0.010	0.517 $\pm$ 0.022	0.115 $\pm$ 0.004	0.925 $\pm$ 0.068	0.6 $\pm$ 0.5	5.686 $\pm$ 2.465	7.051 $\pm$ 0.644	0.395 $\pm$ 0.013	0.885 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Full Data	1.0	0.330 $\pm$ 0.010	0.537 $\pm$ 0.011	0.114 $\pm$ 0.004	0.875 $\pm$ 0.125	1.0 $\pm$ 1.0	4.967 $\pm$ 0.709	8.968 $\pm$ 0.552	0.409 $\pm$ 0.010	0.886 $\pm$ 0.004	8.0 $\pm$ 0.0

Table 2: SMD: method retrained. Distortion and mask-counted payload. Four bytes per value are assumed. RMSE uses standardized values.

Dataset	Method	Budget	MAE	RMSE	NRMSE	Values	Payload bytes	Realized fraction	Max excess values
SMD-1-1	PCA-Triage	0.3	0.258 $\pm$ 0.027	14.902 $\pm$ 7.847	0.016 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Variance	0.3	0.141 $\pm$ 0.002	2.375 $\pm$ 0.415	0.013 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Threshold	0.3	0.413 $\pm$ 0.028	71.459 $\pm$ 11.325	0.019 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Uniform	0.3	0.700 $\pm$ 0.014	116.545 $\pm$ 11.910	0.016 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	PCA-Triage	0.5	0.154 $\pm$ 0.010	9.024 $\pm$ 2.230	0.012 $\pm$ 0.000	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Variance	0.5	0.058 $\pm$ 0.001	1.258 $\pm$ 0.246	0.008 $\pm$ 0.000	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Threshold	0.5	0.109 $\pm$ 0.021	13.987 $\pm$ 17.107	0.013 $\pm$ 0.001	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Uniform	0.5	0.421 $\pm$ 0.084	83.863 $\pm$ 19.805	0.012 $\pm$ 0.000	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	PCA-Triage	0.7	0.039 $\pm$ 0.002	1.902 $\pm$ 0.455	0.006 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Variance	0.7	0.015 $\pm$ 0.000	0.852 $\pm$ 0.065	0.004 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Threshold	0.7	0.021 $\pm$ 0.000	0.562 $\pm$ 0.055	0.004 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Uniform	0.7	0.219 $\pm$ 0.043	55.388 $\pm$ 21.253	0.009 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Full Data	1.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1082202.0 $\pm$ 0.0	4328808.0 $\pm$ 0.0	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000

Table 3: SMD: fixed full data. Detection metrics; delay is conditional and measured in samples.

Dataset	Method	Budget	Precision	Recall	FPR	Event recall	Missed events	Delay (samples)	FA/1000	Point F1	Specificity	Events
SMD-1-1	PCA-Triage	0.3	0.320 $\pm$ 0.016	0.514 $\pm$ 0.019	0.114 $\pm$ 0.006	0.625 $\pm$ 0.000	3.0 $\pm$ 0.0	42.000 $\pm$ 9.445	4.860 $\pm$ 0.319	0.394 $\pm$ 0.016	0.886 $\pm$ 0.006	8.0 $\pm$ 0.0
SMD-1-1	Variance	0.3	0.325 $\pm$ 0.014	0.525 $\pm$ 0.010	0.114 $\pm$ 0.006	0.650 $\pm$ 0.056	2.8 $\pm$ 0.4	25.573 $\pm$ 8.877	5.955 $\pm$ 0.560	0.401 $\pm$ 0.014	0.886 $\pm$ 0.006	8.0 $\pm$ 0.0
SMD-1-1	Threshold	0.3	0.312 $\pm$ 0.011	0.517 $\pm$ 0.013	0.119 $\pm$ 0.006	0.750 $\pm$ 0.088	2.0 $\pm$ 0.7	29.100 $\pm$ 17.772	4.909 $\pm$ 0.248	0.389 $\pm$ 0.011	0.881 $\pm$ 0.006	8.0 $\pm$ 0.0
SMD-1-1	Uniform	0.3	0.320 $\pm$ 0.015	0.516 $\pm$ 0.016	0.115 $\pm$ 0.005	0.675 $\pm$ 0.112	2.6 $\pm$ 0.9	24.406 $\pm$ 9.291	4.242 $\pm$ 0.430	0.395 $\pm$ 0.016	0.885 $\pm$ 0.005	8.0 $\pm$ 0.0
SMD-1-1	PCA-Triage	0.5	0.325 $\pm$ 0.011	0.524 $\pm$ 0.015	0.114 $\pm$ 0.004	0.650 $\pm$ 0.056	2.8 $\pm$ 0.4	27.607 $\pm$ 18.407	6.096 $\pm$ 0.428	0.401 $\pm$ 0.012	0.886 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Variance	0.5	0.328 $\pm$ 0.012	0.532 $\pm$ 0.011	0.114 $\pm$ 0.005	0.900 $\pm$ 0.056	0.8 $\pm$ 0.4	10.450 $\pm$ 9.932	7.486 $\pm$ 0.331	0.406 $\pm$ 0.011	0.886 $\pm$ 0.005	8.0 $\pm$ 0.0
SMD-1-1	Threshold	0.5	0.317 $\pm$ 0.013	0.530 $\pm$ 0.013	0.119 $\pm$ 0.005	0.825 $\pm$ 0.112	1.4 $\pm$ 0.9	11.692 $\pm$ 6.490	7.816 $\pm$ 0.391	0.396 $\pm$ 0.013	0.881 $\pm$ 0.005	8.0 $\pm$ 0.0
SMD-1-1	Uniform	0.5	0.324 $\pm$ 0.011	0.523 $\pm$ 0.011	0.114 $\pm$ 0.005	0.775 $\pm$ 0.056	1.8 $\pm$ 0.4	9.567 $\pm$ 3.312	5.639 $\pm$ 0.304	0.400 $\pm$ 0.010	0.886 $\pm$ 0.005	8.0 $\pm$ 0.0
SMD-1-1	PCA-Triage	0.7	0.328 $\pm$ 0.010	0.531 $\pm$ 0.011	0.114 $\pm$ 0.004	0.775 $\pm$ 0.137	1.8 $\pm$ 1.1	9.450 $\pm$ 6.326	7.606 $\pm$ 0.731	0.406 $\pm$ 0.010	0.886 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Variance	0.7	0.330 $\pm$ 0.011	0.535 $\pm$ 0.012	0.114 $\pm$ 0.004	0.875 $\pm$ 0.125	1.0 $\pm$ 1.0	5.005 $\pm$ 0.805	8.806 $\pm$ 0.712	0.408 $\pm$ 0.011	0.886 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Threshold	0.7	0.328 $\pm$ 0.011	0.535 $\pm$ 0.011	0.115 $\pm$ 0.005	0.825 $\pm$ 0.112	1.4 $\pm$ 0.9	5.718 $\pm$ 1.262	8.722 $\pm$ 0.672	0.407 $\pm$ 0.011	0.885 $\pm$ 0.005	8.0 $\pm$ 0.0
SMD-1-1	Uniform	0.7	0.327 $\pm$ 0.009	0.530 $\pm$ 0.009	0.114 $\pm$ 0.004	0.875 $\pm$ 0.088	1.0 $\pm$ 0.7	5.632 $\pm$ 1.850	6.854 $\pm$ 0.580	0.404 $\pm$ 0.008	0.886 $\pm$ 0.004	8.0 $\pm$ 0.0
SMD-1-1	Full Data	1.0	0.330 $\pm$ 0.010	0.537 $\pm$ 0.011	0.114 $\pm$ 0.004	0.875 $\pm$ 0.125	1.0 $\pm$ 1.0	4.967 $\pm$ 0.709	8.968 $\pm$ 0.552	0.409 $\pm$ 0.010	0.886 $\pm$ 0.004	8.0 $\pm$ 0.0

Table 4: SMD: fixed full data. Distortion and mask-counted payload. Four bytes per value are assumed. RMSE uses standardized values.

Dataset	Method	Budget	MAE	RMSE	NRMSE	Values	Payload bytes	Realized fraction	Max excess values
SMD-1-1	PCA-Triage	0.3	0.258 $\pm$ 0.027	14.902 $\pm$ 7.847	0.016 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Variance	0.3	0.141 $\pm$ 0.002	2.375 $\pm$ 0.415	0.013 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Threshold	0.3	0.413 $\pm$ 0.028	71.459 $\pm$ 11.325	0.019 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Uniform	0.3	0.700 $\pm$ 0.014	116.545 $\pm$ 11.910	0.016 $\pm$ 0.000	324660.0 $\pm$ 0.0	1298640.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	PCA-Triage	0.5	0.154 $\pm$ 0.010	9.024 $\pm$ 2.230	0.012 $\pm$ 0.000	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Variance	0.5	0.058 $\pm$ 0.001	1.258 $\pm$ 0.246	0.008 $\pm$ 0.000	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Threshold	0.5	0.109 $\pm$ 0.021	13.987 $\pm$ 17.107	0.013 $\pm$ 0.001	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Uniform	0.5	0.421 $\pm$ 0.084	83.863 $\pm$ 19.805	0.012 $\pm$ 0.000	541101.0 $\pm$ 0.0	2164404.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	PCA-Triage	0.7	0.039 $\pm$ 0.002	1.902 $\pm$ 0.455	0.006 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Variance	0.7	0.015 $\pm$ 0.000	0.852 $\pm$ 0.065	0.004 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Threshold	0.7	0.021 $\pm$ 0.000	0.562 $\pm$ 0.055	0.004 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Uniform	0.7	0.219 $\pm$ 0.043	55.388 $\pm$ 21.253	0.009 $\pm$ 0.000	757541.0 $\pm$ 0.0	3030164.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
SMD-1-1	Full Data	1.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1082202.0 $\pm$ 0.0	4328808.0 $\pm$ 0.0	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000

Table 5: PSM: method retrained. Detection metrics; delay is conditional and measured in samples.

Dataset	Method	Budget	Precision	Recall	FPR	Event recall	Missed events	Delay (samples)	FA/1000	Point F1	Specificity	Events
PSM	PCA-Triage	0.3	0.992 $\pm$ 0.011	0.015 $\pm$ 0.001	0.000 $\pm$ 0.000	0.094 $\pm$ 0.025	65.2 $\pm$ 1.8	826.403 $\pm$ 131.663	0.023 $\pm$ 0.027	0.030 $\pm$ 0.002	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Variance	0.3	0.985 $\pm$ 0.012	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.122 $\pm$ 0.023	63.2 $\pm$ 1.6	678.380 $\pm$ 299.926	0.025 $\pm$ 0.019	0.032 $\pm$ 0.002	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Threshold	0.3	0.974 $\pm$ 0.024	0.017 $\pm$ 0.002	0.000 $\pm$ 0.000	0.103 $\pm$ 0.027	64.6 $\pm$ 1.9	813.958 $\pm$ 274.543	0.046 $\pm$ 0.037	0.033 $\pm$ 0.004	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Uniform	0.3	0.974 $\pm$ 0.030	0.016 $\pm$ 0.000	0.000 $\pm$ 0.000	0.103 $\pm$ 0.021	64.6 $\pm$ 1.5	683.684 $\pm$ 137.859	0.034 $\pm$ 0.032	0.031 $\pm$ 0.001	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	PCA-Triage	0.5	0.976 $\pm$ 0.031	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.139 $\pm$ 0.026	62.0 $\pm$ 1.9	906.807 $\pm$ 359.620	0.046 $\pm$ 0.046	0.031 $\pm$ 0.001	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Variance	0.5	0.987 $\pm$ 0.014	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.153 $\pm$ 0.022	61.0 $\pm$ 1.6	952.049 $\pm$ 214.155	0.036 $\pm$ 0.025	0.032 $\pm$ 0.001	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Threshold	0.5	0.980 $\pm$ 0.012	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.125 $\pm$ 0.028	63.0 $\pm$ 2.0	985.411 $\pm$ 323.244	0.064 $\pm$ 0.042	0.032 $\pm$ 0.002	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Uniform	0.5	0.974 $\pm$ 0.022	0.017 $\pm$ 0.002	0.000 $\pm$ 0.000	0.147 $\pm$ 0.032	61.4 $\pm$ 2.3	678.599 $\pm$ 402.486	0.068 $\pm$ 0.055	0.033 $\pm$ 0.004	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	PCA-Triage	0.7	0.980 $\pm$ 0.016	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.167 $\pm$ 0.017	60.0 $\pm$ 1.2	742.202 $\pm$ 405.506	0.073 $\pm$ 0.040	0.032 $\pm$ 0.001	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Variance	0.7	0.939 $\pm$ 0.054	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.183 $\pm$ 0.012	58.8 $\pm$ 0.8	692.545 $\pm$ 110.209	0.098 $\pm$ 0.064	0.032 $\pm$ 0.002	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Threshold	0.7	0.980 $\pm$ 0.022	0.018 $\pm$ 0.005	0.000 $\pm$ 0.000	0.181 $\pm$ 0.040	59.0 $\pm$ 2.9	503.929 $\pm$ 136.598	0.109 $\pm$ 0.147	0.036 $\pm$ 0.009	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Uniform	0.7	0.987 $\pm$ 0.013	0.016 $\pm$ 0.001	0.000 $\pm$ 0.000	0.150 $\pm$ 0.033	61.2 $\pm$ 2.4	676.210 $\pm$ 295.402	0.048 $\pm$ 0.047	0.032 $\pm$ 0.002	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Full Data	1.0	0.963 $\pm$ 0.026	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.214 $\pm$ 0.016	56.6 $\pm$ 1.1	629.185 $\pm$ 95.826	0.155 $\pm$ 0.096	0.036 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0

Table 6: PSM: method retrained. Distortion and mask-counted payload. Four bytes per value are assumed. RMSE uses standardized values.

Dataset	Method	Budget	MAE	RMSE	NRMSE	Values	Payload bytes	Realized fraction	Max excess values
PSM	PCA-Triage	0.3	0.109 $\pm$ 0.000	0.290 $\pm$ 0.009	0.017 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Variance	0.3	0.087 $\pm$ 0.000	0.255 $\pm$ 0.027	0.017 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Threshold	0.3	0.101 $\pm$ 0.001	0.281 $\pm$ 0.009	0.019 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Uniform	0.3	0.105 $\pm$ 0.000	0.297 $\pm$ 0.012	0.015 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	PCA-Triage	0.5	0.069 $\pm$ 0.000	0.222 $\pm$ 0.011	0.013 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Variance	0.5	0.045 $\pm$ 0.000	0.193 $\pm$ 0.011	0.013 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Threshold	0.5	0.062 $\pm$ 0.000	0.228 $\pm$ 0.014	0.016 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Uniform	0.5	0.069 $\pm$ 0.000	0.237 $\pm$ 0.018	0.012 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	PCA-Triage	0.7	0.033 $\pm$ 0.000	0.145 $\pm$ 0.010	0.010 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Variance	0.7	0.015 $\pm$ 0.000	0.161 $\pm$ 0.034	0.008 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Threshold	0.7	0.021 $\pm$ 0.000	0.131 $\pm$ 0.014	0.008 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Uniform	0.7	0.039 $\pm$ 0.000	0.176 $\pm$ 0.013	0.009 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Full Data	1.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	2196025.0 $\pm$ 0.0	8784100.0 $\pm$ 0.0	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000

Table 7: PSM: fixed full data. Detection metrics; delay is conditional and measured in samples.

Dataset	Method	Budget	Precision	Recall	FPR	Event recall	Missed events	Delay (samples)	FA/1000	Point F1	Specificity	Events
PSM	PCA-Triage	0.3	0.973 $\pm$ 0.020	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.122 $\pm$ 0.021	63.2 $\pm$ 1.5	967.344 $\pm$ 270.657	0.050 $\pm$ 0.030	0.035 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Variance	0.3	0.982 $\pm$ 0.013	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.150 $\pm$ 0.032	61.2 $\pm$ 2.3	996.798 $\pm$ 258.641	0.039 $\pm$ 0.024	0.035 $\pm$ 0.008	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Threshold	0.3	0.980 $\pm$ 0.015	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.117 $\pm$ 0.025	63.6 $\pm$ 1.8	1311.094 $\pm$ 265.481	0.050 $\pm$ 0.045	0.035 $\pm$ 0.008	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Uniform	0.3	0.970 $\pm$ 0.026	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.122 $\pm$ 0.030	63.2 $\pm$ 2.2	958.219 $\pm$ 280.034	0.055 $\pm$ 0.041	0.035 $\pm$ 0.008	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	PCA-Triage	0.5	0.970 $\pm$ 0.023	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.144 $\pm$ 0.021	61.6 $\pm$ 1.5	1053.611 $\pm$ 302.410	0.080 $\pm$ 0.071	0.035 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Variance	0.5	0.974 $\pm$ 0.018	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.164 $\pm$ 0.021	60.2 $\pm$ 1.5	914.783 $\pm$ 154.680	0.089 $\pm$ 0.058	0.035 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Threshold	0.5	0.972 $\pm$ 0.018	0.018 $\pm$ 0.003	0.000 $\pm$ 0.000	0.136 $\pm$ 0.050	62.2 $\pm$ 3.6	852.212 $\pm$ 339.140	0.096 $\pm$ 0.067	0.034 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Uniform	0.5	0.971 $\pm$ 0.028	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.150 $\pm$ 0.023	61.2 $\pm$ 1.6	789.026 $\pm$ 199.579	0.061 $\pm$ 0.061	0.035 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	PCA-Triage	0.7	0.969 $\pm$ 0.024	0.018 $\pm$ 0.003	0.000 $\pm$ 0.000	0.181 $\pm$ 0.026	59.0 $\pm$ 1.9	758.742 $\pm$ 54.960	0.112 $\pm$ 0.065	0.035 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Variance	0.7	0.969 $\pm$ 0.024	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.183 $\pm$ 0.021	58.8 $\pm$ 1.5	723.110 $\pm$ 142.210	0.116 $\pm$ 0.079	0.035 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Threshold	0.7	0.967 $\pm$ 0.027	0.018 $\pm$ 0.003	0.000 $\pm$ 0.000	0.192 $\pm$ 0.021	58.2 $\pm$ 1.5	719.205 $\pm$ 119.058	0.121 $\pm$ 0.080	0.036 $\pm$ 0.006	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Uniform	0.7	0.974 $\pm$ 0.026	0.018 $\pm$ 0.003	0.000 $\pm$ 0.000	0.169 $\pm$ 0.015	59.8 $\pm$ 1.1	762.339 $\pm$ 170.357	0.093 $\pm$ 0.072	0.035 $\pm$ 0.006	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0
PSM	Full Data	1.0	0.963 $\pm$ 0.026	0.018 $\pm$ 0.004	0.000 $\pm$ 0.000	0.214 $\pm$ 0.016	56.6 $\pm$ 1.1	629.185 $\pm$ 95.826	0.155 $\pm$ 0.096	0.036 $\pm$ 0.007	1.000 $\pm$ 0.000	72.0 $\pm$ 0.0

Table 8: PSM: fixed full data. Distortion and mask-counted payload. Four bytes per value are assumed. RMSE uses standardized values.

Dataset	Method	Budget	MAE	RMSE	NRMSE	Values	Payload bytes	Realized fraction	Max excess values
PSM	PCA-Triage	0.3	0.109 $\pm$ 0.000	0.290 $\pm$ 0.009	0.017 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Variance	0.3	0.087 $\pm$ 0.000	0.255 $\pm$ 0.027	0.017 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Threshold	0.3	0.101 $\pm$ 0.001	0.281 $\pm$ 0.009	0.019 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Uniform	0.3	0.105 $\pm$ 0.000	0.297 $\pm$ 0.012	0.015 $\pm$ 0.000	658807.0 $\pm$ 0.0	2635228.0 $\pm$ 0.0	0.300 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	PCA-Triage	0.5	0.069 $\pm$ 0.000	0.222 $\pm$ 0.011	0.013 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Variance	0.5	0.045 $\pm$ 0.000	0.193 $\pm$ 0.011	0.013 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Threshold	0.5	0.062 $\pm$ 0.000	0.228 $\pm$ 0.014	0.016 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Uniform	0.5	0.069 $\pm$ 0.000	0.237 $\pm$ 0.018	0.012 $\pm$ 0.000	1098012.0 $\pm$ 0.0	4392048.0 $\pm$ 0.0	0.500 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	PCA-Triage	0.7	0.033 $\pm$ 0.000	0.145 $\pm$ 0.010	0.010 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Variance	0.7	0.015 $\pm$ 0.000	0.161 $\pm$ 0.034	0.008 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Threshold	0.7	0.021 $\pm$ 0.000	0.131 $\pm$ 0.014	0.008 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Uniform	0.7	0.039 $\pm$ 0.000	0.176 $\pm$ 0.013	0.009 $\pm$ 0.000	1537217.0 $\pm$ 0.0	6148868.0 $\pm$ 0.0	0.700 $\pm$ 0.000	0.000 $\pm$ 0.000
PSM	Full Data	1.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	2196025.0 $\pm$ 0.0	8784100.0 $\pm$ 0.0	1.000 $\pm$ 0.000	0.000 $\pm$ 0.000

Table 9: TEP: method retrained. Detection metrics; delay is conditional and measured in samples.

Dataset	Method	Budget	Precision	Recall	FPR	Event recall	Missed events	Delay (samples)	FA/1000	Point F1	Specificity	Events
TEP	PCA-Triage	0.5	0.963 ± 0.001	0.904 ± 0.005	0.384 ± 0.012	1.000 ± 0.000	0.0 ± 0.0	2.962 ± 0.879	7.223 ± 0.391	0.932 ± 0.002	0.616 ± 0.012	100.0 ± 0.0
TEP	Variance	0.5	0.962 ± 0.001	0.903 ± 0.004	0.393 ± 0.009	1.000 ± 0.000	0.0 ± 0.0	3.460 ± 0.554	7.170 ± 0.668	0.931 ± 0.002	0.607 ± 0.009	100.0 ± 0.0
TEP	Threshold	0.5	0.965 ± 0.002	0.886 ± 0.007	0.354 ± 0.023	1.000 ± 0.000	0.0 ± 0.0	4.588 ± 0.696	5.760 ± 0.398	0.924 ± 0.003	0.646 ± 0.023	100.0 ± 0.0
TEP	Uniform	0.5	0.962 ± 0.001	0.914 ± 0.002	0.399 ± 0.011	1.000 ± 0.000	0.0 ± 0.0	2.044 ± 0.438	9.672 ± 0.148	0.937 ± 0.001	0.601 ± 0.011	100.0 ± 0.0
TEP	Full Data	1.0	0.943 ± 0.002	0.972 ± 0.002	0.638 ± 0.019	1.000 ± 0.000	0.0 ± 0.0	0.350 ± 0.304	8.343 ± 1.114	0.958 ± 0.001	0.362 ± 0.019	100.0 ± 0.0

Table 10: TEP: method retrained. Distortion and mask-counted payload. Four bytes per value are assumed. RMSE uses standardized values.

Dataset	Method	Budget	MAE	RMSE	NRMSE	Values	Payload bytes	Realized fraction	Max excess values
TEP	PCA-Triage	0.5	0.104 ± 0.000	0.270 ± 0.002	0.027 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Variance	0.5	0.099 ± 0.000	0.261 ± 0.002	0.026 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Threshold	0.5	0.139 ± 0.001	0.383 ± 0.002	0.038 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Uniform	0.5	0.175 ± 0.000	0.483 ± 0.001	0.036 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Full Data	1.0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	2730000.0 ± 0.0	10920000.0 ± 0.0	1.000 ± 0.000	0.000 ± 0.000

Table 11: TEP: fixed full data. Detection metrics; delay is conditional and measured in samples.

Dataset	Method	Budget	Precision	Recall	FPR	Event recall	Missed events	Delay (samples)	FA/1000	Point F1	Specificity	Events
TEP	PCA-Triage	0.5	0.940 ± 0.001	0.980 ± 0.001	0.688 ± 0.014	1.000 ± 0.000	0.0 ± 0.0	0.242 ± 0.094	7.878 ± 0.378	0.960 ± 0.001	0.312 ± 0.014	100.0 ± 0.0
TEP	Variance	0.5	0.939 ± 0.001	0.980 ± 0.001	0.691 ± 0.017	1.000 ± 0.000	0.0 ± 0.0	0.242 ± 0.094	7.779 ± 0.274	0.959 ± 0.000	0.309 ± 0.017	100.0 ± 0.0
TEP	Threshold	0.5	0.939 ± 0.001	0.983 ± 0.001	0.698 ± 0.012	1.000 ± 0.000	0.0 ± 0.0	0.242 ± 0.094	7.333 ± 0.146	0.960 ± 0.001	0.302 ± 0.012	100.0 ± 0.0
TEP	Uniform	0.5	0.940 ± 0.001	0.975 ± 0.001	0.678 ± 0.015	1.000 ± 0.000	0.0 ± 0.0	0.242 ± 0.094	8.560 ± 0.149	0.958 ± 0.001	0.322 ± 0.015	100.0 ± 0.0
TEP	Full Data	1.0	0.943 ± 0.002	0.972 ± 0.002	0.638 ± 0.019	1.000 ± 0.000	0.0 ± 0.0	0.350 ± 0.304	8.343 ± 1.114	0.958 ± 0.001	0.362 ± 0.019	100.0 ± 0.0

Table 12: TEP: fixed full data. Distortion and mask-counted payload. Four bytes per value are assumed. RMSE uses standardized values.

Dataset	Method	Budget	MAE	RMSE	NRMSE	Values	Payload bytes	Realized fraction	Max excess values
TEP	PCA-Triage	0.5	0.104 ± 0.000	0.270 ± 0.002	0.027 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Variance	0.5	0.099 ± 0.000	0.261 ± 0.002	0.026 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Threshold	0.5	0.139 ± 0.001	0.383 ± 0.002	0.038 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Uniform	0.5	0.175 ± 0.000	0.483 ± 0.001	0.036 ± 0.000	1365000.0 ± 0.0	5460000.0 ± 0.0	0.500 ± 0.000	0.000 ± 0.000
TEP	Full Data	1.0	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	2730000.0 ± 0.0	10920000.0 ± 0.0	1.000 ± 0.000	0.000 ± 0.000

Table 13: SMD: method retrained. Point F1 at 50% bandwidth across predeclared training-score quantiles; Full Data is the full-rate reference. The primary quantile remains 0.99.

Method	0.95	0.975	0.99	0.995	0.999
PCA-Triage	0.389 ± 0.009	0.423 ± 0.017	0.390 ± 0.017	0.369 ± 0.004	0.355 ± 0.003
Variance	0.375 ± 0.022	0.417 ± 0.019	0.407 ± 0.019	0.390 ± 0.016	0.357 ± 0.004
Threshold	0.392 ± 0.023	0.427 ± 0.018	0.393 ± 0.013	0.371 ± 0.008	0.350 ± 0.001
Uniform	0.377 ± 0.029	0.430 ± 0.014	0.395 ± 0.008	0.373 ± 0.005	0.357 ± 0.002
Full Data	0.383 ± 0.018	0.431 ± 0.027	0.409 ± 0.010	0.388 ± 0.009	0.356 ± 0.002

Table 14: SMD: fixed full data. Point F1 at 50% bandwidth across predeclared training-score quantiles; Full Data is the full-rate reference. The primary quantile remains 0.99.

Method	0.95	0.975	0.99	0.995	0.999
PCA-Triage	0.386 ± 0.019	0.448 ± 0.026	0.401 ± 0.012	0.370 ± 0.002	0.357 ± 0.001
Variance	0.382 ± 0.019	0.433 ± 0.026	0.406 ± 0.011	0.385 ± 0.009	0.356 ± 0.002
Threshold	0.374 ± 0.018	0.429 ± 0.027	0.396 ± 0.013	0.375 ± 0.010	0.349 ± 0.003
Uniform	0.385 ± 0.020	0.446 ± 0.027	0.400 ± 0.010	0.375 ± 0.006	0.356 ± 0.002
Full Data	0.383 ± 0.018	0.431 ± 0.027	0.409 ± 0.010	0.388 ± 0.009	0.356 ± 0.002

Table 15: PSM: method retrained. Point F1 at 50% bandwidth across predeclared training-score quantiles; Full Data is the full-rate reference. The primary quantile remains 0.99.

Method	0.95	0.975	0.99	0.995	0.999
PCA-Triage	0.277 ± 0.047	0.104 ± 0.011	0.031 ± 0.001	0.027 ± 0.000	0.023 ± 0.000
Variance	0.267 ± 0.072	0.098 ± 0.032	0.032 ± 0.001	0.027 ± 0.000	0.023 ± 0.000
Threshold	0.237 ± 0.052	0.088 ± 0.016	0.032 ± 0.002	0.027 ± 0.000	0.024 ± 0.001
Uniform	0.268 ± 0.061	0.102 ± 0.029	0.033 ± 0.004	0.027 ± 0.000	0.023 ± 0.001
Full Data	0.272 ± 0.047	0.108 ± 0.023	0.036 ± 0.007	0.027 ± 0.000	0.024 ± 0.000

Table 16: PSM: fixed full data. Point F1 at 50% bandwidth across predeclared training-score quantiles; Full Data is the full-rate reference. The primary quantile remains 0.99.

Method	0.95	0.975	0.99	0.995	0.999
PCA-Triage	0.269 ± 0.049	0.105 ± 0.021	0.035 ± 0.007	0.027 ± 0.000	0.023 ± 0.000
Variance	0.272 ± 0.049	0.105 ± 0.023	0.035 ± 0.007	0.027 ± 0.000	0.023 ± 0.000
Threshold	0.271 ± 0.052	0.104 ± 0.023	0.034 ± 0.007	0.027 ± 0.000	0.022 ± 0.001
Uniform	0.270 ± 0.048	0.105 ± 0.022	0.035 ± 0.007	0.027 ± 0.000	0.023 ± 0.000
Full Data	0.272 ± 0.047	0.108 ± 0.023	0.036 ± 0.007	0.027 ± 0.000	0.024 ± 0.000